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Project Title:	Dynamic Response of Particle-Filled Polymeric Fluids in Flows with Complex Lagrangian Time Histories		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
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	Particle-filled polymeric fluids, frequently encountered in processing composite materials into fibers or sheets, turbulent drag reduction, and hydraulic fracturing, undergo alternating periods of simple shear and extensional motion in a Lagrangian reference frame. The dynamic rheological response of a particle-polymer mixture is not a simple superposition of independent polymeric and particle stresses. Instead, there is significant coupling between the velocity disturbance produced by the particles, polymer stress, and particle orientation. There is a dearth of relevant experiments and theoretical/ numerical analyses exploring the dynamic rheology of these systems. The proposed project provides an opportunity to employ theoretical and numerical simulation methods uniquely suited to predict the particle-induced polymer stress in transient flows and validate these methods by comparison with transient extensional rheology experiments of particle-polymer mixtures subject to pre-shearing in the Physical Sciences Informatics (PSI) database.		

Task Description:	A filament stretching rheometer used in the Shear History Extensional Rheology Experiments (SHERE, SHERE II, SHERE-Reflight / SHERE-R) enabled the measurement of a particle-filled polymeric fluid's transient extensional flow response subject to variable pre-shearing rates. SHERE and SHERE-R provide complementary measurements for the same fluid without particles. The experiments use dilute particle suspensions in a well-characterized Boger fluid over a range of moderate extension rates, conditions unachievable in Earth's gravity. A statistical analysis of the SHERE data indicates a significant reduction in strain hardening by particles at high strains, which agrees with a preliminary theoretical analysis. Prediction of the coupled particle and polymer stresses is arduous because the particle-induced velocity disturbance greatly alters the polymer stress in regions close to the particle surface, leading to strong polymer stress gradients that are difficult to resolve numerically and subtle to predict theoretically. The interplay between particle orientation, velocity disturbance, and polymer stresses when fibers are suspended in polymeric fluids presents further challenges. The project will employ theory and numerical simulations developed in our group, uniquely suited to address these problems. A finite difference method will solve the inertia-less viscoelastic flow in a spheroidal coordinate system that conforms to and rotates with the surface of a spheroidal particle and applies a time-varying linear velocity boundary condition at a nearly spherical outer boundary far from the particle. A demonstration that the ensemble average of the linear perturbation to the polymer stress due to the particle velocity disturbance is zero is essential to obtain convergent bulk rheological properties from numerical simulations. A complementary theoretical approach will yield semi-analytical predictions of the suspension rheology valid in the limit of small polymer concentration, c. Comparison with the numerical simulation will reveal the range of c, over which this theory is quantitatively accurate, and provide useful qualitative predictions for larger c. The theory is less computationally intensive and avoids computational stability limitations typical of nonlinear viscoelastic flow computations at high Deborah numbers and polymer extensibilities. Hence, it facilitates exploring broader parameter regimes. These methods will be applied to study the rheology of (1) suspensions of spheres in transient extensional flow with and without preshearing for comparison with the SHERE experiments; (2) suspensions of spheroids experiencing a simple shear flow preceded and followed by a period of extensional flow to model the processing of fiber-filled polymeric films; and (3) suspensions of spheroids subject to simple shear flow alternating with short bursts of extensional motion as a model of turbulent drag reduction by the coupled effects of fiber and polymer additives.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Suspensions of particles within polymeric or viscoelastic fluids are integral to numerous manufacturing and industrial processes. Within these applications, these suspensions navigate intricate flows, ranging from shear caused by solid surfaces to extension through orifices. The purpose behind introducing particles, such as spheres or fibers, is often to confer specific attributes upon the final product, such as heightened mechanical strength or directional electrical conductivity. However, the interaction between particles and polymers within the fluid is a pivotal factor dictating the energy demands throughout the manufacturing process. This interaction significantly alters the rheological properties of the suspension in comparison to the particle-free polymeric fluid. This study provides quantitative predictions and mechanistic understanding of the effects of particle-polymer interactions on the flow resistance of such materials.
	Suspensions of particles within polymeric or viscoelastic fluids are integral to numerous manufacturing and industrial processes. Within these applications, these suspensions navigate intricate flows, ranging from shear caused by solid surfaces to extension through orifices. The purpose behind introducing particles, such as spheres or fibers, is often to confer specific attributes upon the final product, such as heightened mechanical strength or directional electrical conductivity. However, the interaction between particles and polymers within the fluid is a pivotal factor dictating the energy demands throughout the manufacturing process. This interaction significantly alters the rheological properties of the suspension in comparison to the particle-free polymeric fluid. Presently, cutting-edge studies in fluid dynamics depict polymers as springs affixed to spherical beads. This model effectively captures qualitative phenomena, including the well-known coil-stretch transition. Nevertheless, achieving precise quantitative modeling and experimental characterization of polymers remains an ongoing challenge. The complexity arises from the diverse chemical compositions of polymers and their varied responses to external factors like temperature fluctuations. Consequently, beyond merely quantifying the impact of particle-polymer interaction on suspension rheology, it is imperative to delve into the underlying mechanisms at play. Understanding these intricate dynamics is paramount for advancing both theoretical models and practical applications in industrial settings. We employ a combination of theoretical techniques and a robust numerical approach to delve into the interaction between particles and polymers in dilute particle suspensions in polymeric fluids. Our theoretical work is particularly relevant for low polymer concentrations, enabling us to explore a wide spectrum of parameters, including polymer relaxation time and particle aspect ratio. A key ingredient of our theory is the use of a generalized reciprocal theorem to obtain the effect of polymers on fluid velocity and pressure without resorting to computationally expensive methods. This approach allows us to scrutinize extreme scenarios effectively. Our numerical method, developed in prolate spheroidal coordinates with the particle as one of the coordinate surfaces, facilitates the study of diverse particle shapes, ranging from spheres to moderately elongated prolate spheroids. Importantly, this method extends the reach of our theoretical findings to encompass higher polymer concentrations. It accommodates various polymer constitutive models, enabling us to provide mechanistic explanations applicable to an array of fluids, from dilute polymer solutions to melts of entangled polymers. Through this numerical method, we can explore different types of flows, including simple shear, uniaxial extensional, or combinations of linear flows that vary with time. Our computational domain consists of the particle surface on the inside and a far-field nearly spherical outer boundary. Several features of the chosen coordinate system, such as the natural clustering of the grid near the particle surface, allow us to consider a wider range of parameters, such as particle aspect ratio and polymer relaxation time, compared to previous numerical attempts. This is possible because the region near the particle surface consists of large polymer gradients that can be efficiently captured using the prolate spheroidal grid. This comprehensive approach allows us to gain profound insights into the behavior of particle-polymer suspensions across a wide spectrum of conditions and fluid types.

Task Progress:

Since we are dealing with dilute particle suspensions, interactions between particles are rare. The net effect of particles in a suspension is an average of the ensembles of particles in the fluid flow under consideration. Thus, the flow around an isolated particle in an unbounded fluid is sufficient to characterize the suspension rheology. Some previous studies have simply replaced the ensemble average with a volume average. However, this approach leads to mathematical divergence of the integrals involving the polymer stress. Our recent theoretical construct of linearized polymer stress allows us to overcome this issue. This stress is due to a polymer constitutive model obtained by linearizing it about the imposed flow. While the volume average of this linearized stress causes the aforementioned divergence, its ensemble average is zero. Thus, before approximating the ensemble average with the volume average, we remove the linearized polymer stress. This not only provides a theoretical guarantee for convergence but also allows our numerical solution to converge within a reasonable value of the outer radius of the computational domain. In our rheological calculations, the interaction between particles and polymers manifests as a combination of the interaction stresslet and the particle-induced polymer stress (PIPS). The former represents the additional stress due to interactions in the particle phase, while the latter signifies the extra polymer stress created by the flow disturbance caused by the particle.

The interaction between particles and polymers significantly affects the extensional rheology of a suspension of spheres in polymeric liquids, both qualitatively and quantitatively. This effect stems from the coil-stretch transition of polymers undisturbed by particles. When subjected to strain with large polymer relaxation times, polymers shift from a coiled to a fully stretched state, leading to a rapid increase in polymer stress. At low polymer concentrations, c, during the extension of the sphere suspension, polymers downstream of the particles form regions of higher stretch than the undisturbed far-field stretch. With increased strain or over time, these stretched polymer regions extend downstream due to the finite memory of polymer relaxation time, creating a wake of highly stretched polymers and increasing the suspension stress, primarily through the particle-induced polymer stress (PIPS). In cases of small polymer relaxation time, this pattern stabilizes into a steady state with positive interaction stress. However, with larger relaxation times, the far-field polymers eventually undergo the coil-stretch transition. Hence the polymers near particles are less extended than the highly stretched ones in the far-field. This deficit in polymer stretch results in negative PIPS. Consequently, at steady state, in suspensions with large polymer relaxation times, the extensional stress is lower than in a particle-free fluid.

At low polymer concentrations (small c), the influence of the polymers' stretch on the underlying flow field is limited. However, with an increase in c, regions of high polymer stretch resist the velocity gradient, causing the stretch to be resisted, while regions of low polymer stretch enhance the velocity gradient, enhancing the stretch. The most significant changes in extensional rheology due to increasing c occur in cases with larger polymer relaxation times, both at steady state and in long duration transient flows when the net particle-polymer interaction stress is negative. In this scenario, increasing c weakens the collapsed polymer regions near particles. This happens because the collapsed polymers act as a low-viscosity fluid region, enabling previously collapsed polymers at low c to stretch more at higher c. Initially, there is a slight reduction in the magnitude of the negative PIPS (normalized with c) upon increasing c. However, with further increase in c, the c-normalized PIPS (and consequently the net interaction stress) becomes more negative due to a new mechanism activated in a wider fluid region surrounding the particle. At low c, polymers approaching the particle from the far-field were slightly more stretched than their undisturbed value in this region. However, due to increased feedback between polymer stretch and the underlying fluid velocity field, this region becomes one of slightly collapsed polymers. This broader area of slightly less stretched polymers results in an increasingly negative (c-normalized) PIPS as c increases. The general observation at larger polymer relaxation time of a positive interaction stress when the extensional flow first starts followed by a negative effect of particles is qualitatively observed in the SHERE experimental dataset available in the PSI database.

When a polymeric material flows near walls, like ducts and pipes in manufacturing processes, it undergoes localized simple shear flow. To validate the ability of our computational method to describe the rheology of particle filled polymeric liquids in shear flows, we simulated a start up shear flow of dilute suspension of spheres in a polymer melt at a moderately high polymer concentration and compared the results with the work of Jain and Shaqfeh (J. Rheol. 65, 1269 (2021)). We obtained good quantitative agreement with the previous simulations at low to moderate polymer relaxation times and qualitative agreement at higher polymer relaxation times. Specifically, both studies showed that the contribution of the PIPS to the suspension viscosity increased with increasing shear rate while the particle stresslet contribution shear thinned. We confirmed that our simulations were well resolved with respect to grid size, domain size and time stepping even at the highest polymer relaxation times.

Anisotropic particles can be used to impart large anisotropic changes in the electrical and mechanical properties of a composite material. Since the fluid stress arising during melt processing of such materials is strongly dependent on the particle orientation, an understanding of the particle-polymer interaction's effect on particle orientation dynamics is a precursor to characterizing the shear rheology of such suspensions. Using prolate spheroids as prototypical anisotropic particles, we investigate the impact of polymer concentration, relaxation time, and particle aspect ratio on these dynamics. In a Newtonian fluid, these particles follow Jeffery orbits, which are initial-orientation-dependent non-uniform periodic orbits around the vorticity axis of the imposed shear flow. Viscoelasticity, or the influence of polymers, breaks this degeneracy. At low polymer concentrations (c), we establish a two-dimensional phase diagram in the space of particle aspect ratio versus the product of polymer relaxation time and c, denoted as cDe. At low cDe, particles tend to align nearly with the vorticity axis, settling in smaller periodic orbits. Conversely, high cDe causes particles to stabilize in a steady state aligned with the imposed flow direction. At intermediate cDe values, the final orientation depends on the particle's initial orientation. At larger values of c, depending on the other parameters, we observe either one or two steady state orientations in the flow-vorticity plane, between the flow and vorticity directions. The ability to obtain a steady state orientation in a polymeric fluid, instead of degenerate Jeffery orbits in Newtonian fluid, provides better control to achieve a uniform particle orientation that is vital in applications such as manufacturing of anisotropic thin films for flexible and transparent electronics.

So far, our studies have focused on characterizing the rheology of sphere suspensions in polymeric fluid under purely shear or purely extensional flow. Our next steps involve studying these suspensions in a time-varying imposed flow inspired by practical applications like fiber spinning. In this process, the suspension is initially sheared within the spinneret and then pulled by the drawing mechanism, creating a strong uniaxial extensional flow before solidifying into a fiber. Additionally, after understanding the influence of polymers on the orientation of anisotropic particles like prolate spheroids in simple shear flow, we are strategically designing a simulation study to explore the shear rheology of fiber suspensions in polymeric fluids. This will be essential for understanding the energy requirements in processes such as roll-to-roll manufacturing, where anisotropic particles are used to introduce anisotropy into thin films.

Bibliography Type:	Description: (Last Updated: 10/15/2025)
Articles in Other Journals or Periodicals	Sharma A, Koch DL. "Rotation of a fibre in simple shear flow of a dilute polymer solution." Journal of Fluid Mechanics. 2023 Dec 28;976:A9. Submitted. http://dx.doi.org/10.1017/jfm.2023.823 , Dec-2023
Articles in Peer-reviewed Journals	Sharma A, Koch DL. "Steady-state extensional rheology of a dilute suspension of spheres in a dilute polymer solution." Phys. Rev. Fluids. 2023 Mar;8(3):033303. https://doi.org/10.1103/PhysRevFluids.8.033303 , Mar-2023